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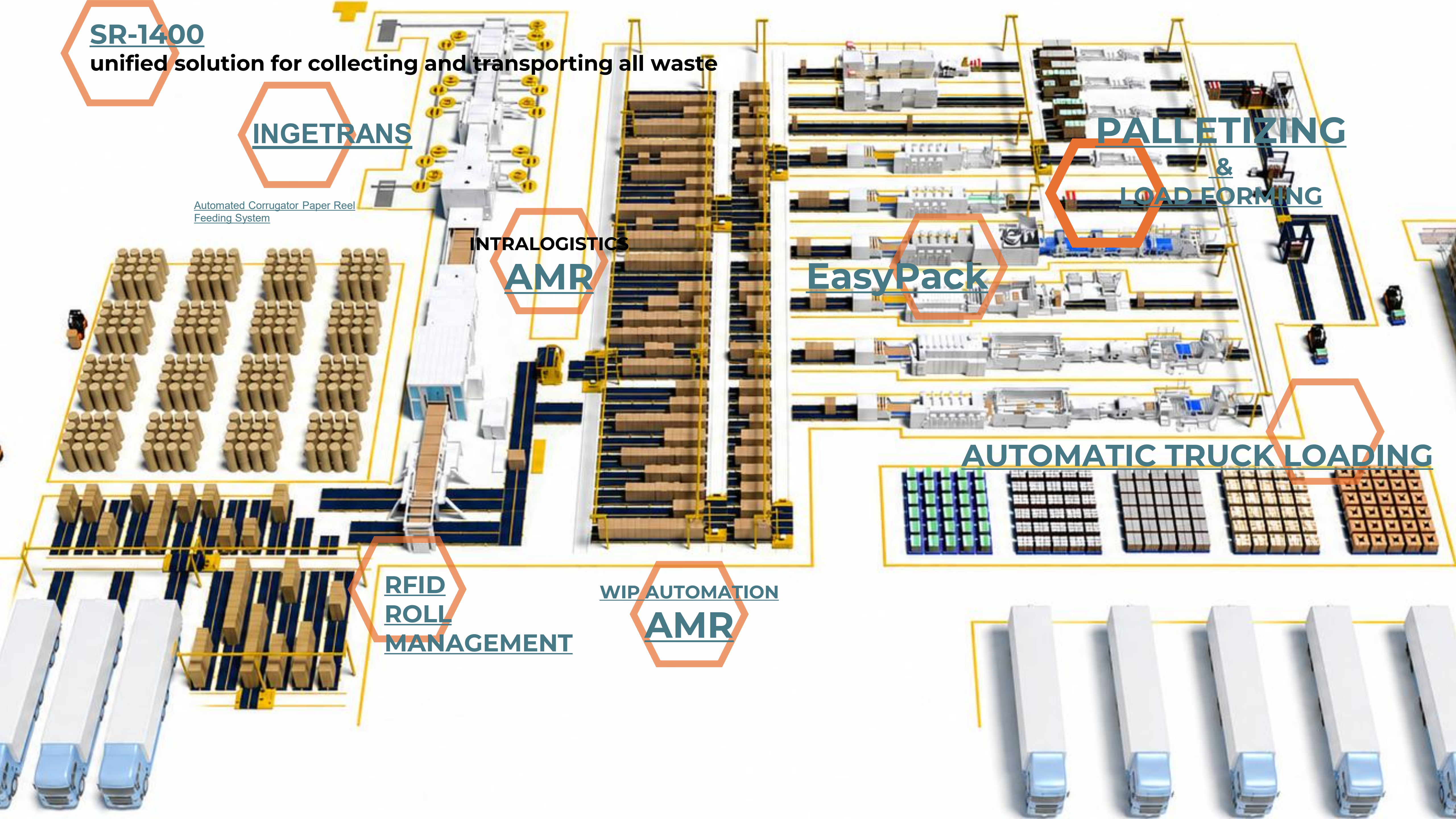
ENGINEERING & AUDITING

TAILORED AUTOMATION
& INTRALOGISTICS

The Corrugated Plant of the Future Starts Here

Corrugated Plant Automation Solutions


SR-1400
unified solution for collecting and transporting all waste
INGETRANS
Automated Corrugator Paper Reel Feeding System
INTRALOGISTICS
AMR
EasyPack
PALLETIZING
&
LOAD FORMING
AUTOMATIC TRUCK LOADING
RFID
ROLL
MANAGEMENT
WIP AUTOMATION
AMR

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TAILORED AUTOMATION
& INTRALOGISTICS

1.268	28	26	194
Projects	Years	Agreements	Installations

WHAT WE DO

Industrial machinery	Operational efficiency and reduced manual handling
Installation & commissioning	Seamless integration into your existing plant flow
Robotics & palletizing	Lower error rates, higher throughput, improved safety
Project management	Single point of contact for the entire project lifecycle
360° turnkey solution	Design, execution, training, and startup from A to Z
Technical auditing	Independent diagnosis, free from manufacturer bias
Industrial Intelligence (Ing_PRO)	AI applied to operations, maintenance, and quality

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TAILORED AUTOMATION & INTRALOGISTICS

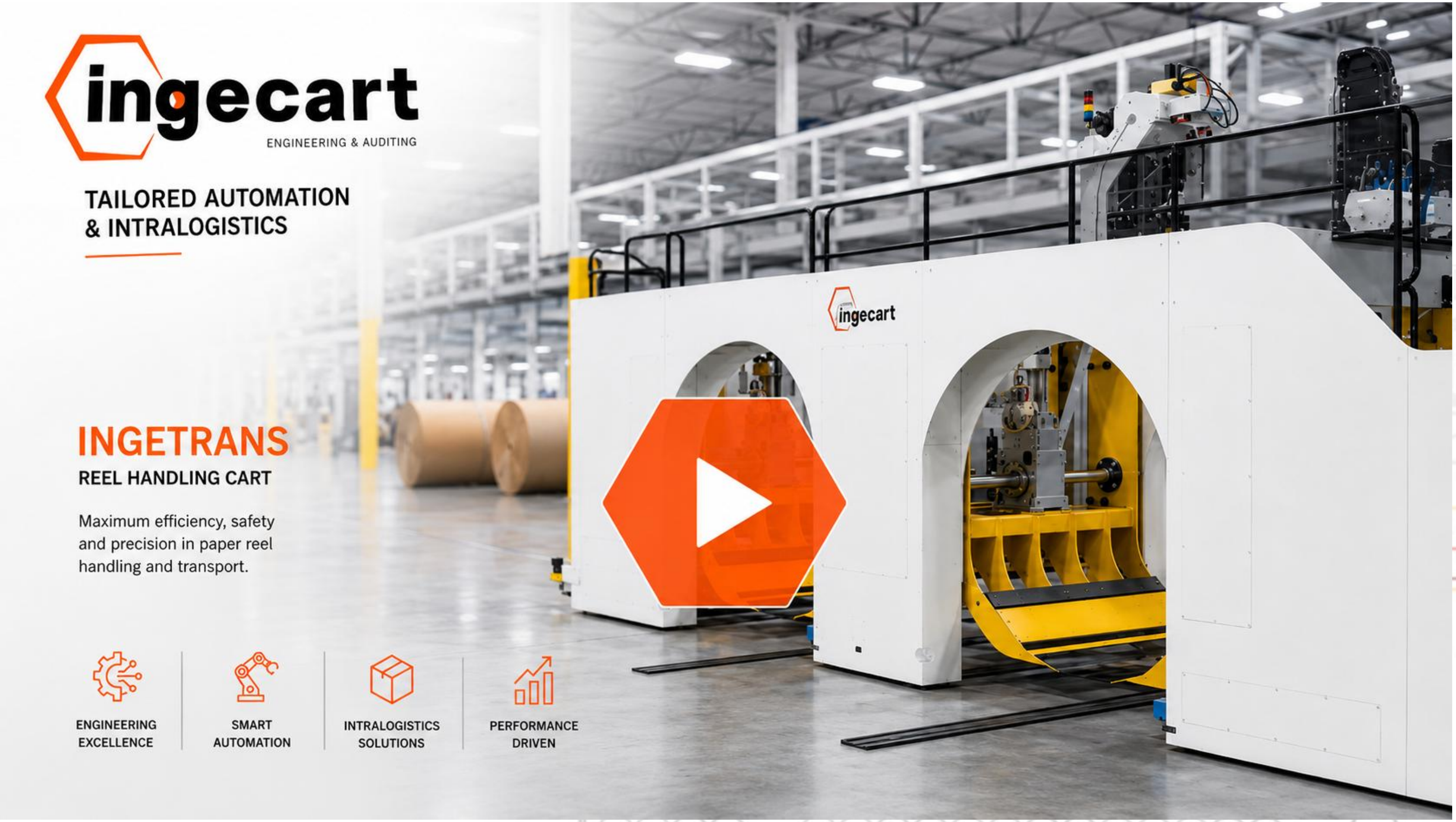
INGETRANS

REEL HANDLING CART

Maximum efficiency, safety and precision in paper reel handling and transport.

ENGINEERING EXCELLENCE | SMART AUTOMATION | INTRALOGISTICS SOLUTIONS | PERFORMANCE DRIVEN

INTRALOGISTICS INGETRANS:



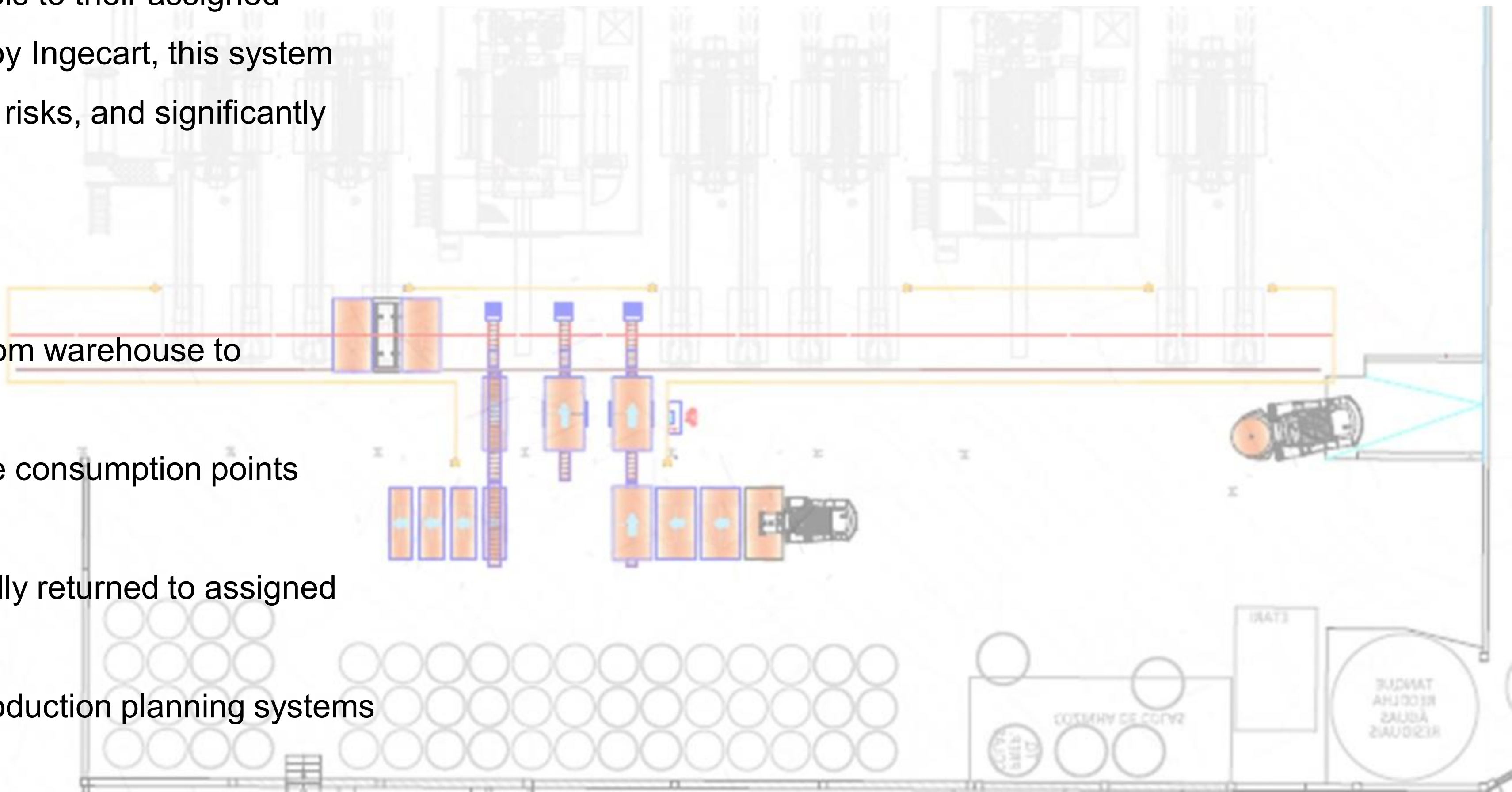
INTRALOGISTICS INGETRANS

Automated Corrugator Paper Reel Feeding System

INGETRANS is a complete automated solution for managing paper reel supply to multiple consumption points on the corrugator, including the automatic return of unused, or partially used, reels to their assigned storage locations. Designed and manufactured by Ingecart, this system eliminates forklift dependency, reduces accident risks, and significantly improves operational efficiency.

KEY FUNCTIONS

- Automated movement: Rells are transported from warehouse to corrugator without manual intervention
- Automated delivery: Precise delivery to multiple consumption points (single facers, roll stands)
- Automated return: Unused rolls are automatically returned to assigned warehouse locations
- Process integration: Works seamlessly with production planning systems



INTRALOGISTICS INGETRANS

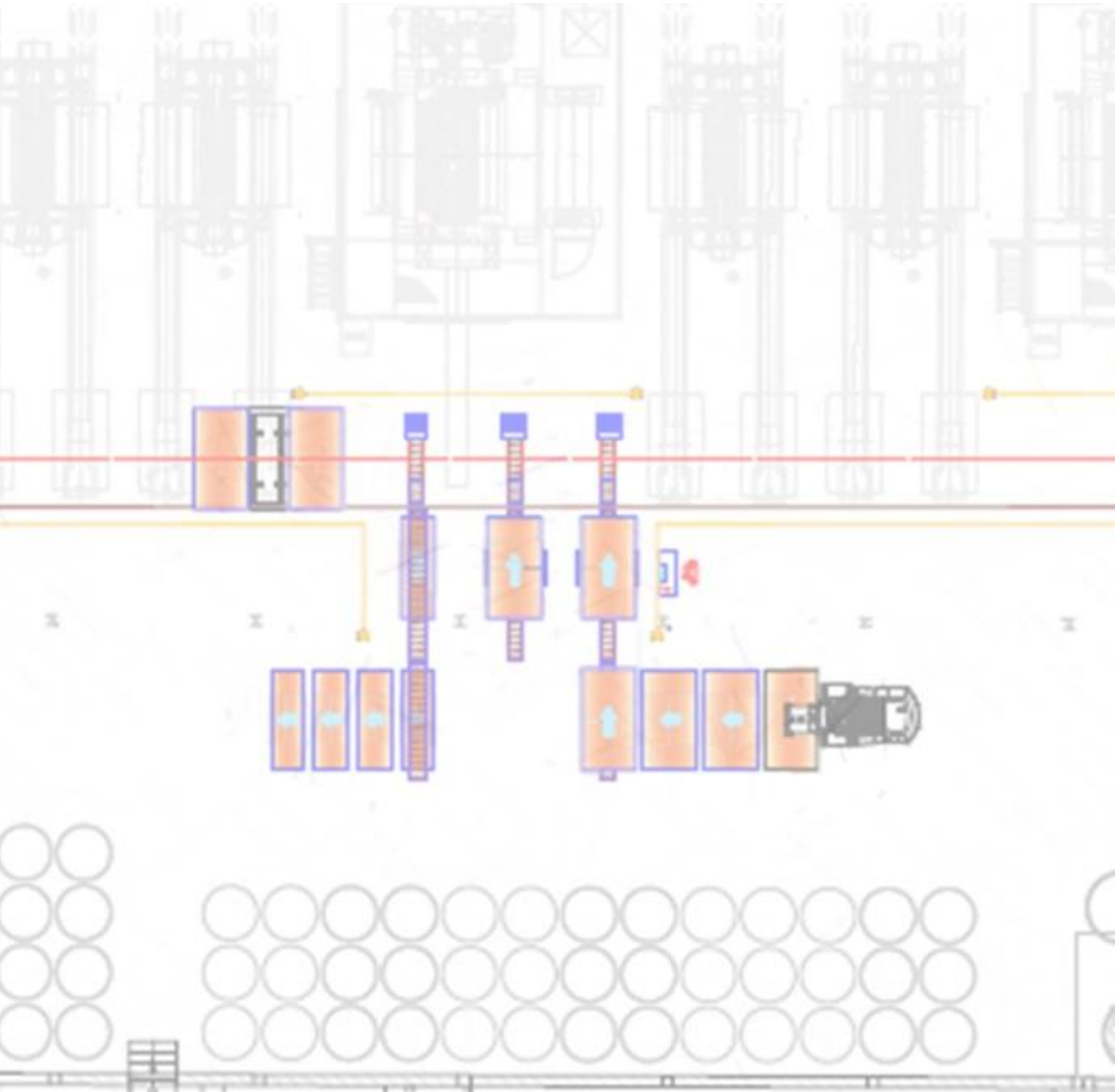
Ingetrans– Corrugator Relel Change Cycle Time Calculation

Calculation basis: Delivery and return cycle for 5 rolls, Minimum Order Length at 300 m/min

Description	Seconds	Minutes
Total cycle time – Delivery & return of 5 rolls	346.72	**5.78**
Average time per roll	69.34	1.16

Parameter	Value	Unit
Corrugator speed	300	m/min
Minimum order length at 300 m/min	**1,733.6**	**meters**

INTRALOGISTICS INGETRANS



TECHNICAL SPECIFICATIONS

- System type: Surface rail-guided vehicle
- Track system: Floor-mounted rails
- Operation: Fully automated (no forklift or direct personnel intervention)
- Adaptability: Custom design per client layout (number of lanes, length, consumption points)
- Integration: Compatible with customer's planning / MES system

BENEFITS

- Eliminates forklift traffic – removes uncontrolled vehicle movement from corrugator work area
- Reduces accident risk – protects personnel operating near machines
- Labor cost reduction – minimizes manual roll handling tasks
- Minimizes downtime – rolls arrive exactly when needed
- Fast ROI – quick payback through efficiency gains and cost savings
- Increased profitability – higher overall equipment effectiveness (OEE)

HOW IT WORKS

1. Start – The process begins in the customer's warehouse
2. Transport – The Ingetrans carriage moves along floor-mounted rails
3. Delivery – Rolls are delivered to consumption points on the corrugator
4. Return – Unused rolls are automatically returned to assigned locations

More meters per minute.
Zero chaos at the corrugator.

Safe productivity for corrugated plants.

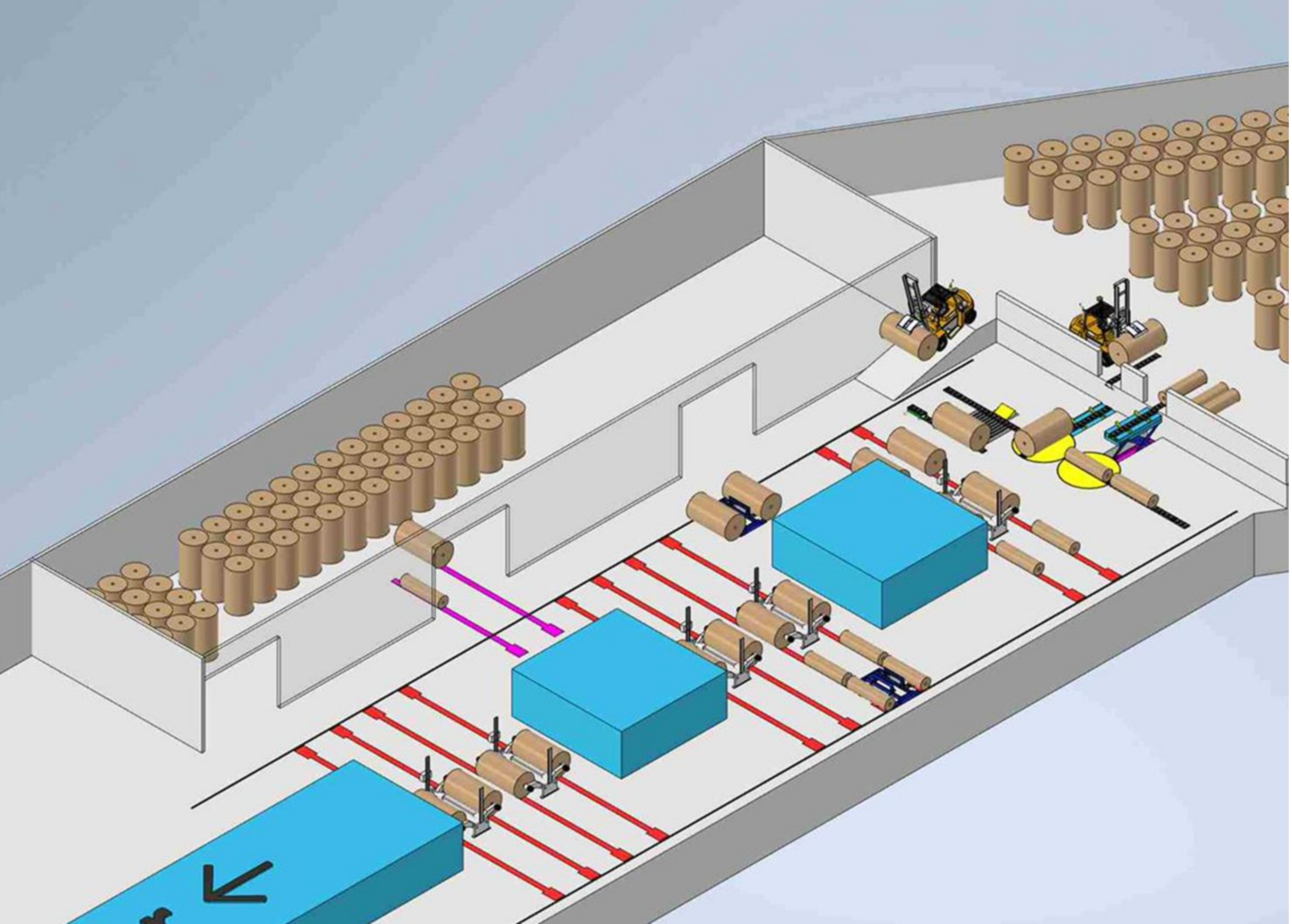
More real line capacity: continuous, production-synchronized supply.

Automates reel movement, delivery, and return from warehouse to corrugator.

Creates safer workspaces by eliminating uncontrolled circulation.

The Corrugated Plant of the **Future**

...Possible layouts, real examples



Case #1

two loading positions → Work in Progress and Main warehouse

Two cars; loading and back to warehouse

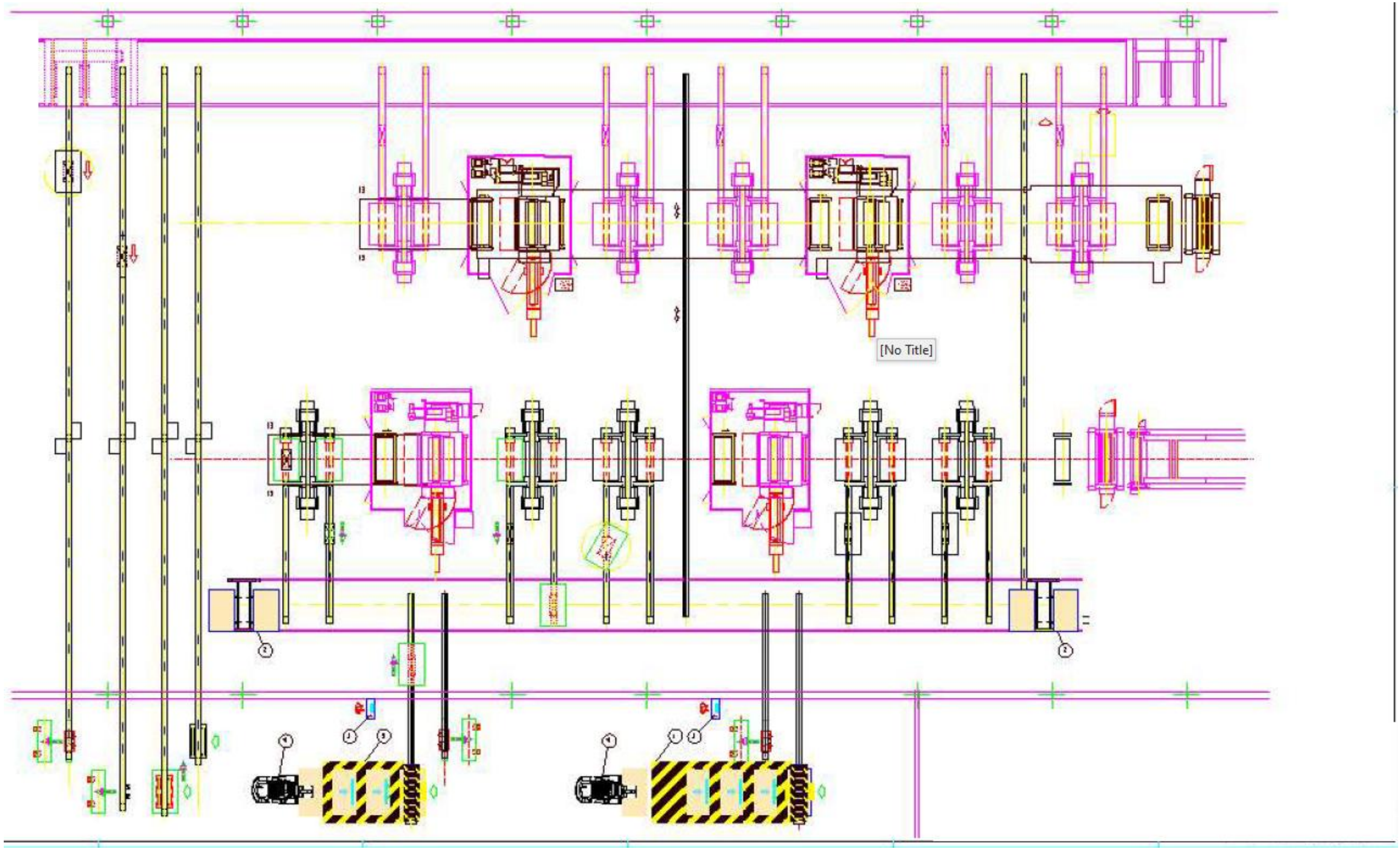
“A” Project

Item	Cant.	Descripción
1	1	Rampa con Alamcenamiento de 3 bobinas de 2800 mm
2	2	Transfer-Car Doble sobre suelo para Bobinas de 2800 mm
3	2	Cuadro Principal de Mando
4	2	Pantalla en Carretilla Elevadora
1	1	Rampa con Alamcenamiento de 2 bobinas de 2800 mm

Calculo Ciclo entrega 6 Bobinas 2 Transfer		
	Seg	**Min**
Entrega y Retorno 6 Bobinas	**197,09**	3,28
Tiempo medio Bobina	32,85	0,55

Pedido mínimo a 300 mt/min		
Velocidad Onduladora	300	mt/min
Tiempo cambio Bobinas	3,28	min
Pedido mínimo a 300 mt/min	985,435	mt

“A” Project



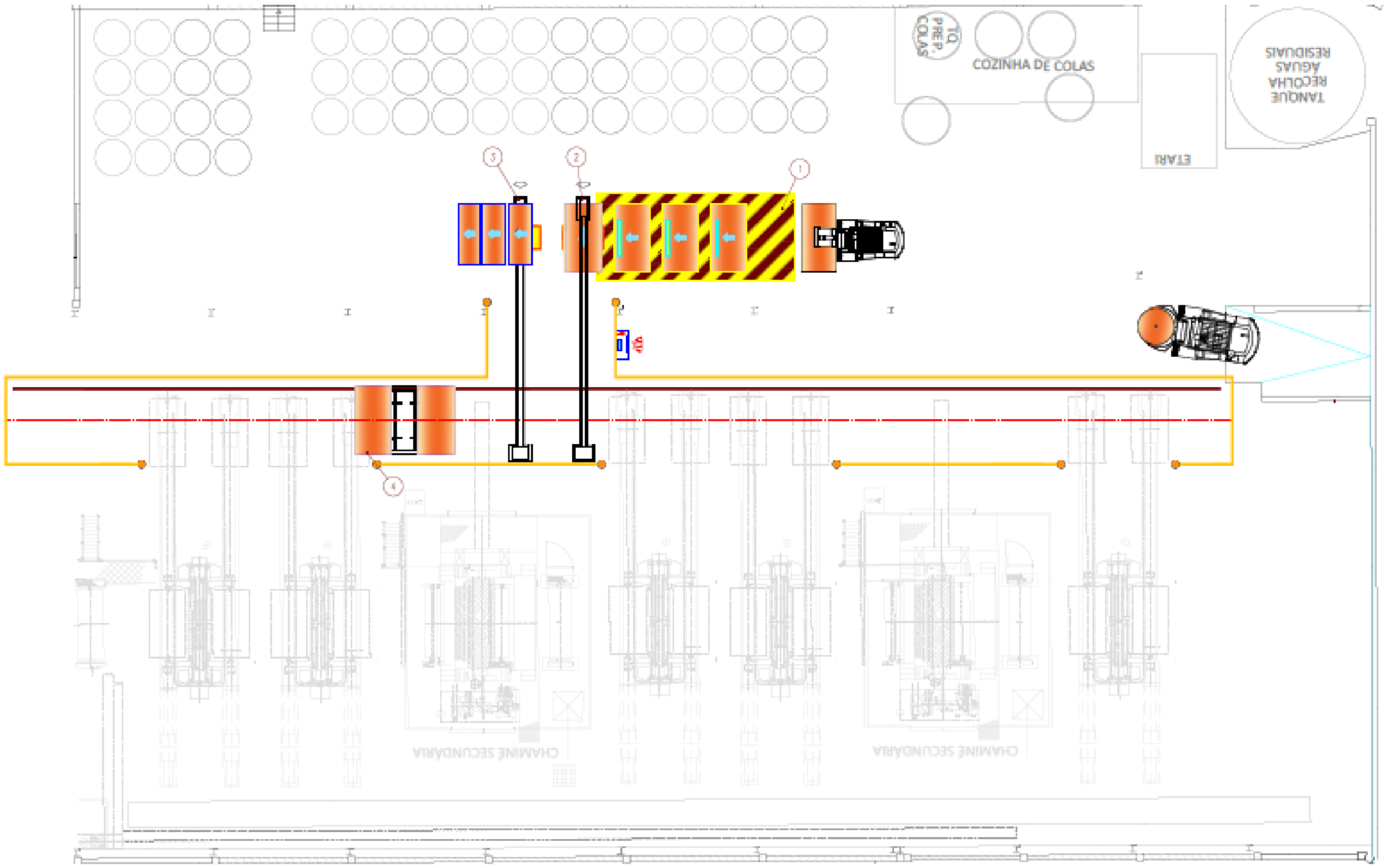
“DSM” Project

Item	Cant.	Descripción
1	1	Rampa con Alamcenamiento de 3 bobinas
2	1	Carril Estandar Motorizado de 11 mt. con 2 Floor Flaps
3	1	Carril Estandar Motorizado de 11 mt. con Expulsor
4	1	Transfer-Car Doble sobre suelo

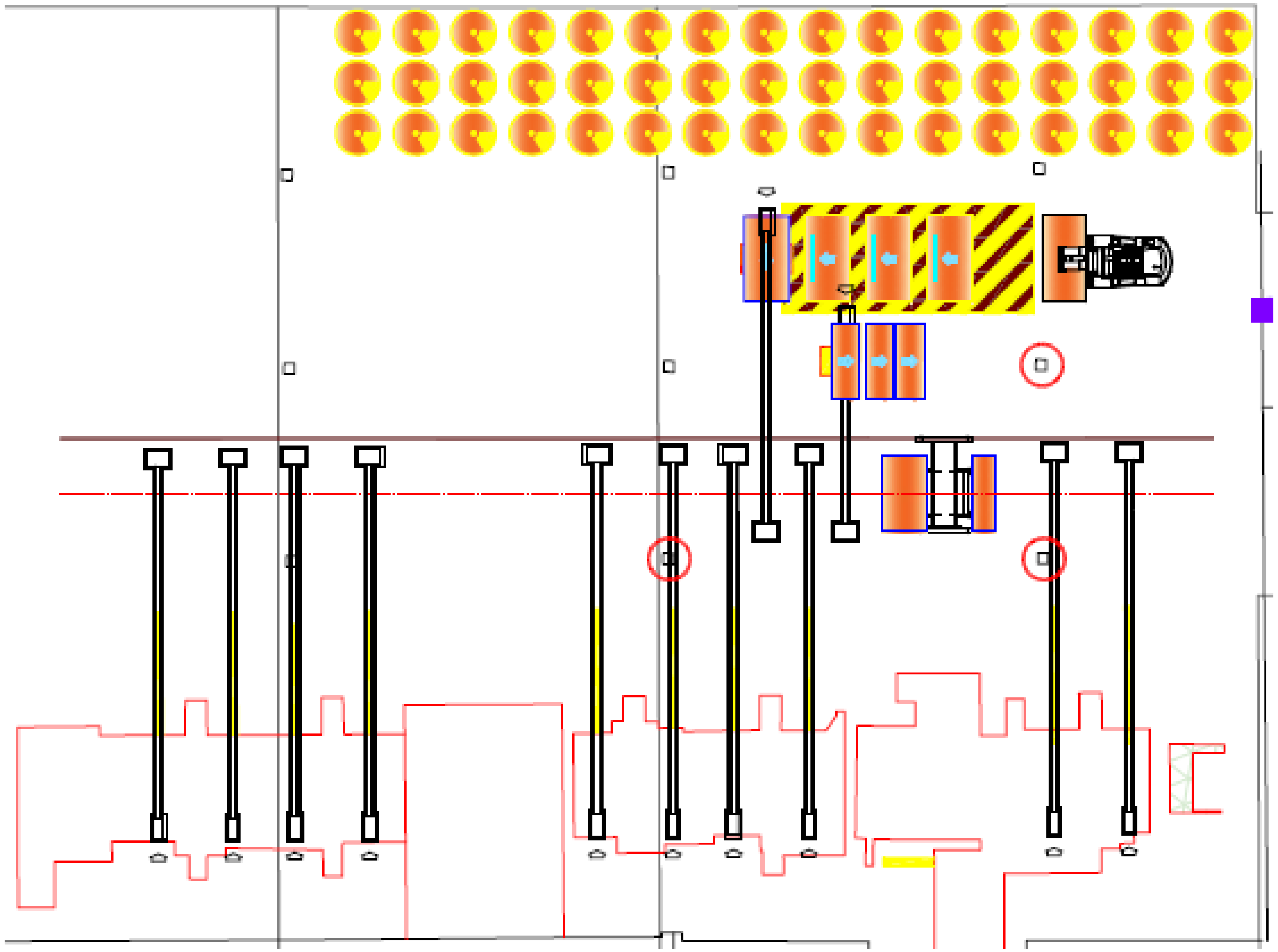
Entrega 5 Bobinas y 5 de Retorno		
	Seg	Min
Tiempo Entrega 6 Bobinas	**486,25**	8,10
Tiempo medio Bobina	81,04	1,35

Pedido mínimo a 300 mt/min		
Velocidad Onduladora	300	mt/min
Tiempo cambio Bobinas	8,1	min
Pedido mínimo a 300 mt/min	2430	mt

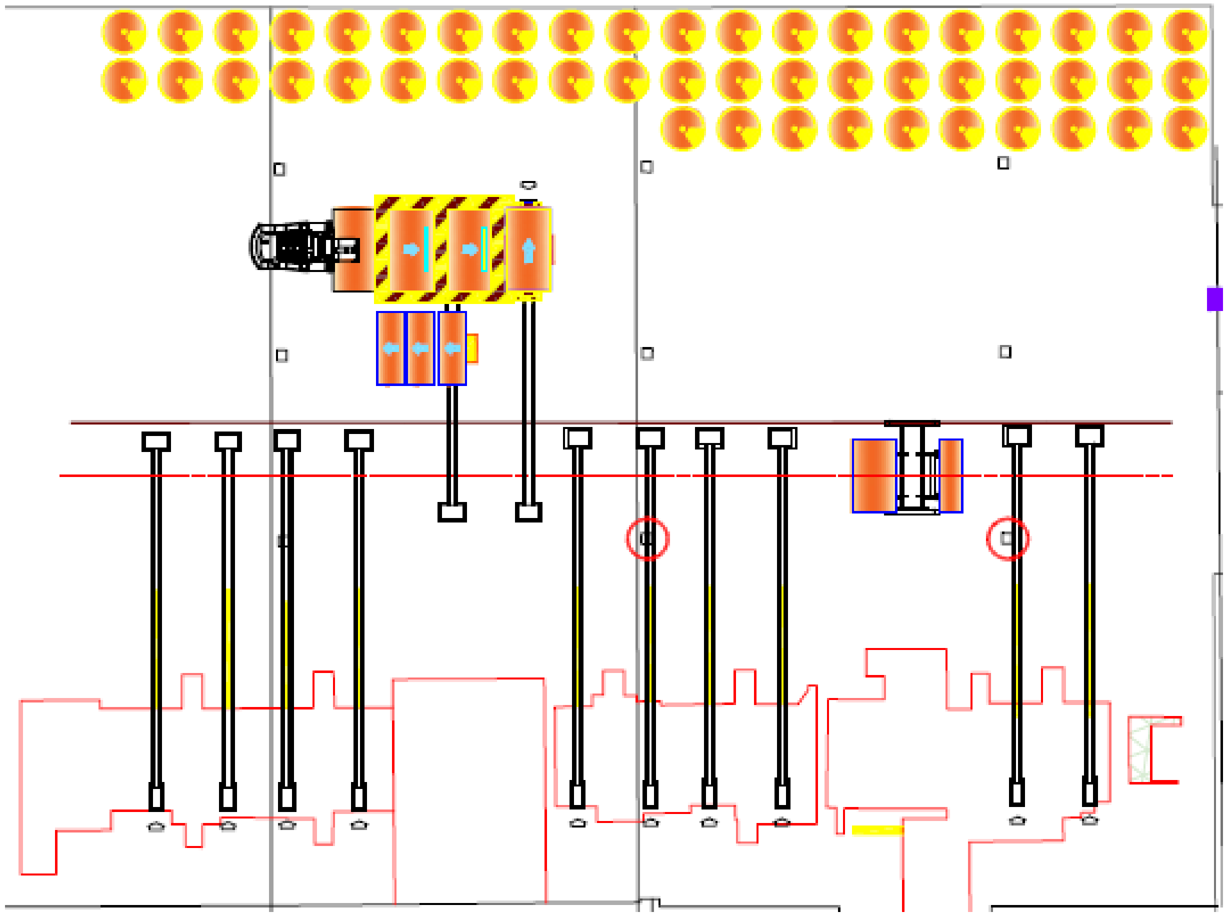
“DSM” Project



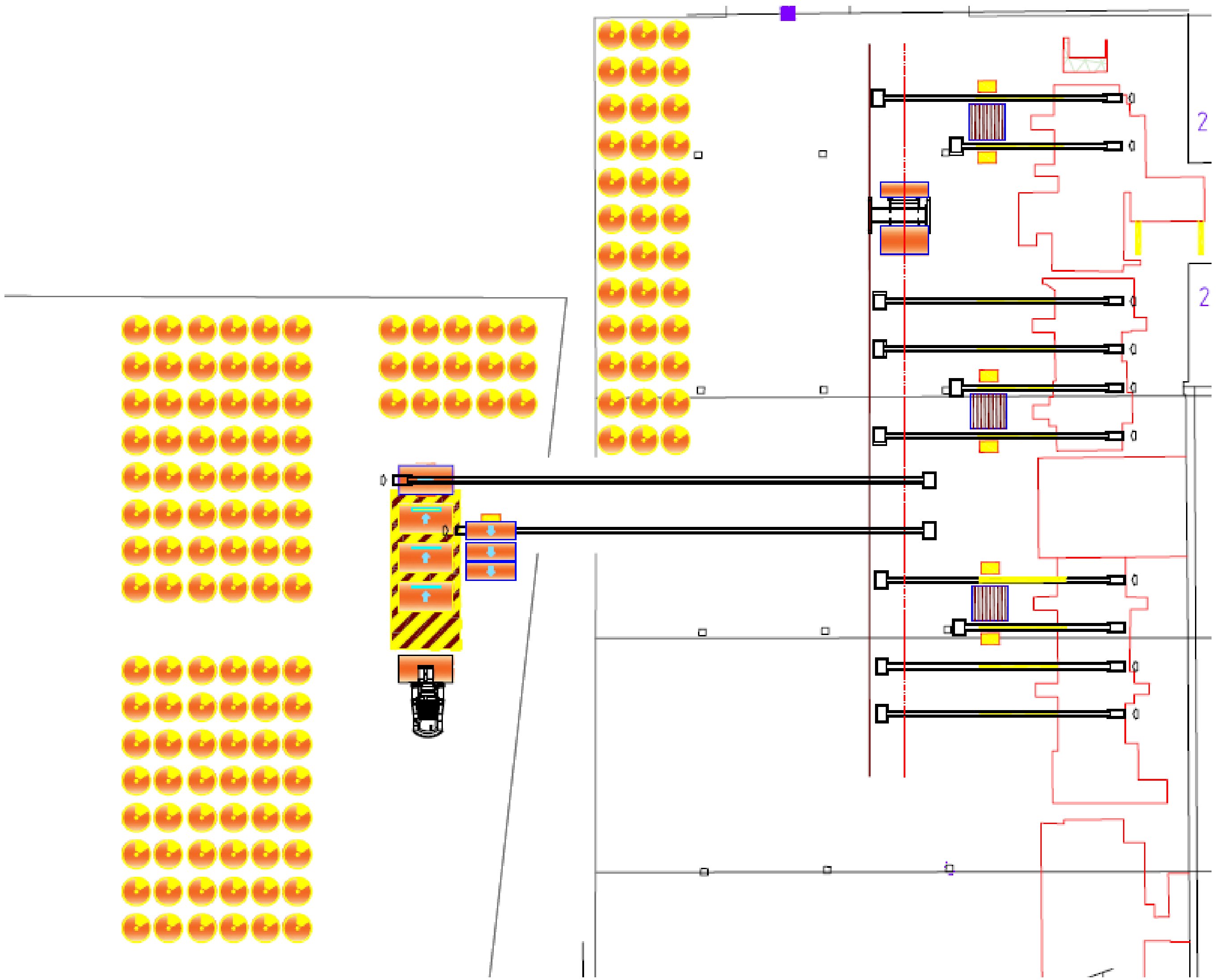
"CV" Project



CV-2



CV-3



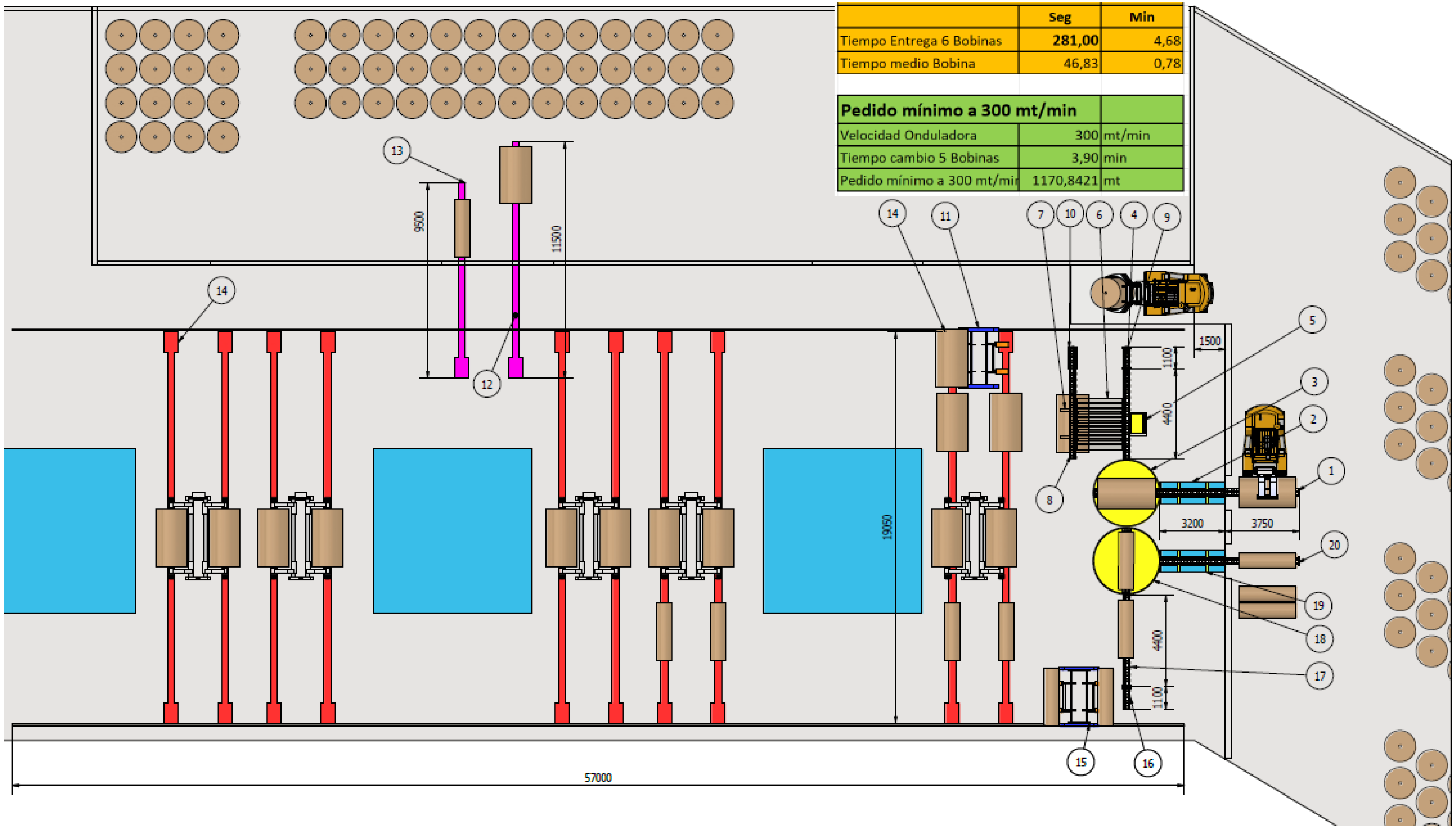
M1230

Lista de Elementos			Lista de Elementos		
Ref.	Cant.	Descripción	Ref.	Cant.	Descripción
1	1	Carril de Cadena 3,75 mt.	11	1	Transfercar Doble con Rail de 57 mt.
2	1	Tijera Hidráulica Elevadora con Carril de Cadena de 3,2 mt.	12	1	Carril con 1 Carro 11,5 mt.
3	1	Carril Giratorio 3,2 mt.	13	1	Carril con 1 Carro 9,5 mt.
4	1	Carril de Cadena 4,4 mt.	14	10	Carril con 2 Carros 19 mt.
5	1	Empujador	15	1	Transfercar Simple con Rail de 57 mt.
6	1	Transportador Neumático	16	1	Carril de Cadena 1,1 mt. con Elevación Hidráulica.
7	1	Tope de Goma	17	1	Carril de Cadena 4,4 mt.
8	1	Carril de Cadena 4,4 mt.	18	1	Carril Giratorio 3,2 mt.
9	1	Carril de Cadena 1,1 mt. con Elevación Hidráulica.	19	1	Tijera Hidráulica Elevadora con Carril de Cadena de 3,2 mt.
10	1	Carril de Cadena 1,1 mt. con Elevación Hidráulica.	20	1	Carril de Cadena 3,75 mt. con Cilindro Expulsor hidráulico

Cálculo Ciclo entrega 6 Bobinas		
	Seg	Min
Tiempo Entrega 6 Bobinas	281,00	4,68
Tiempo medio Bobina	46,83	0,78

Pedido mínimo a 300 mt/min		
Velocidad Onduladora	300	mt/min
Tiempo cambio 5 Bobinas	3,90	min
Pedido mínimo a 300 mt/mir	1170,8421	mt

M1230



	Seg	Min
Tiempo Entrega 6 Bobinas	**281,00**	4,68
Tiempo medio Bobina	46,83	0,78

Pedido mínimo a 300 mt/min		
Velocidad Onduladora	300	mt/min
Tiempo cambio 5 Bobinas	3,90	min
Pedido mínimo a 300 mt/mi	1170,8421	mt

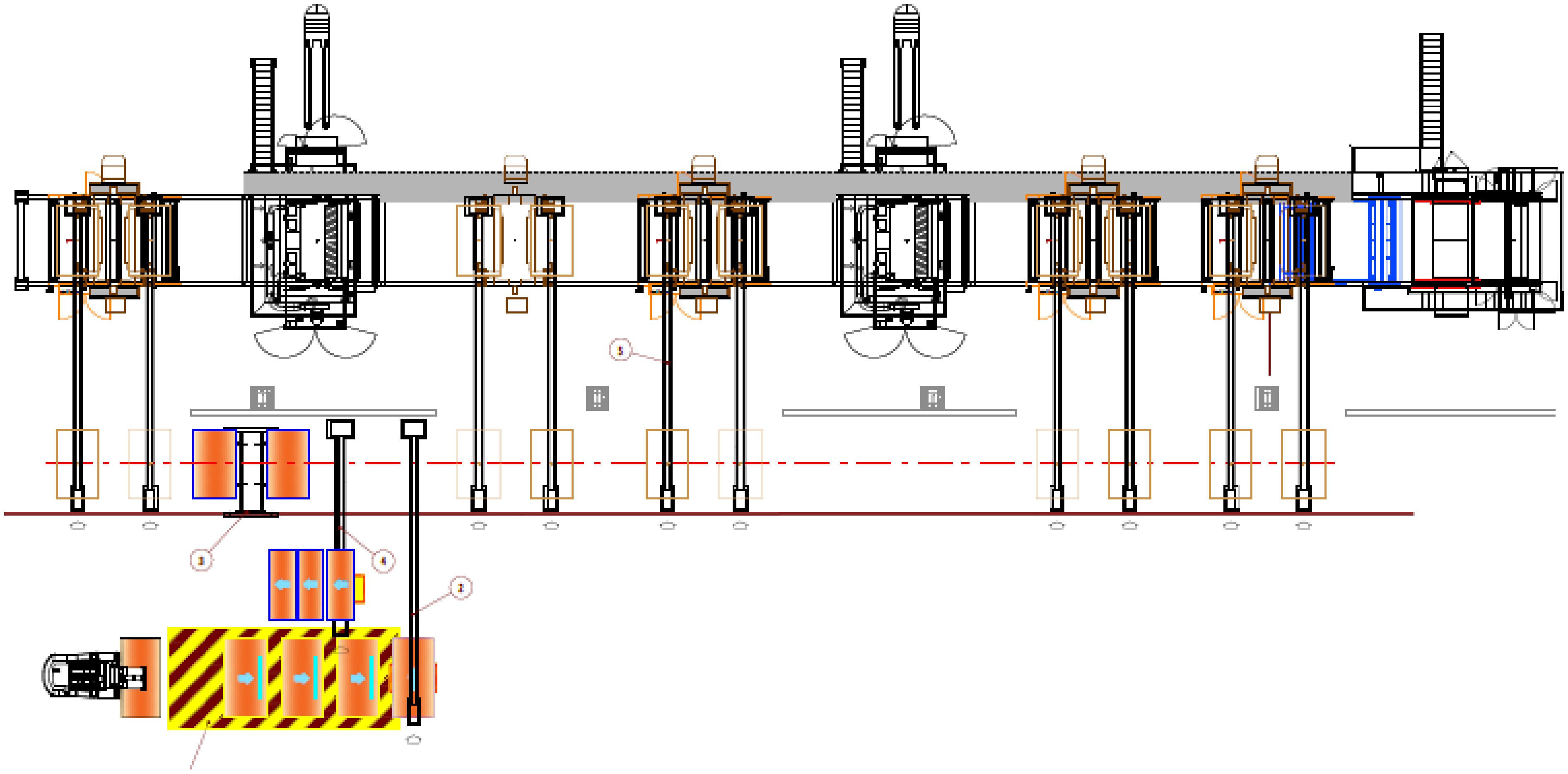
“TP” Project

Item	Cant.	Descripción
1	1	Rampa con Alamcenamiento de 3 bobinas
2	1	Carril Estandar de 11 mt. con Floor Flap
3	1	Transfer-Car Doble sobre suelo
4	1	Carril Estándar de 8 mt. con Expulsor
5	10	Carril Estándar de 11,5 mt. con carro motorizado

Calculo Ciclo entrega 5 Bobinas		
	Seg	Min
Entrega y Retorno 5 Bobinas	**335,85**	5,60
Tiempo medio Bobina	67,17	1,12

Pedido mínimo a 300 mt/min		
Velocidad Onduladora	300	mt/min
Tiempo cambio Bobinas	5,60	min
Pedido mínimo a 300 mt/min	1679,25	mt

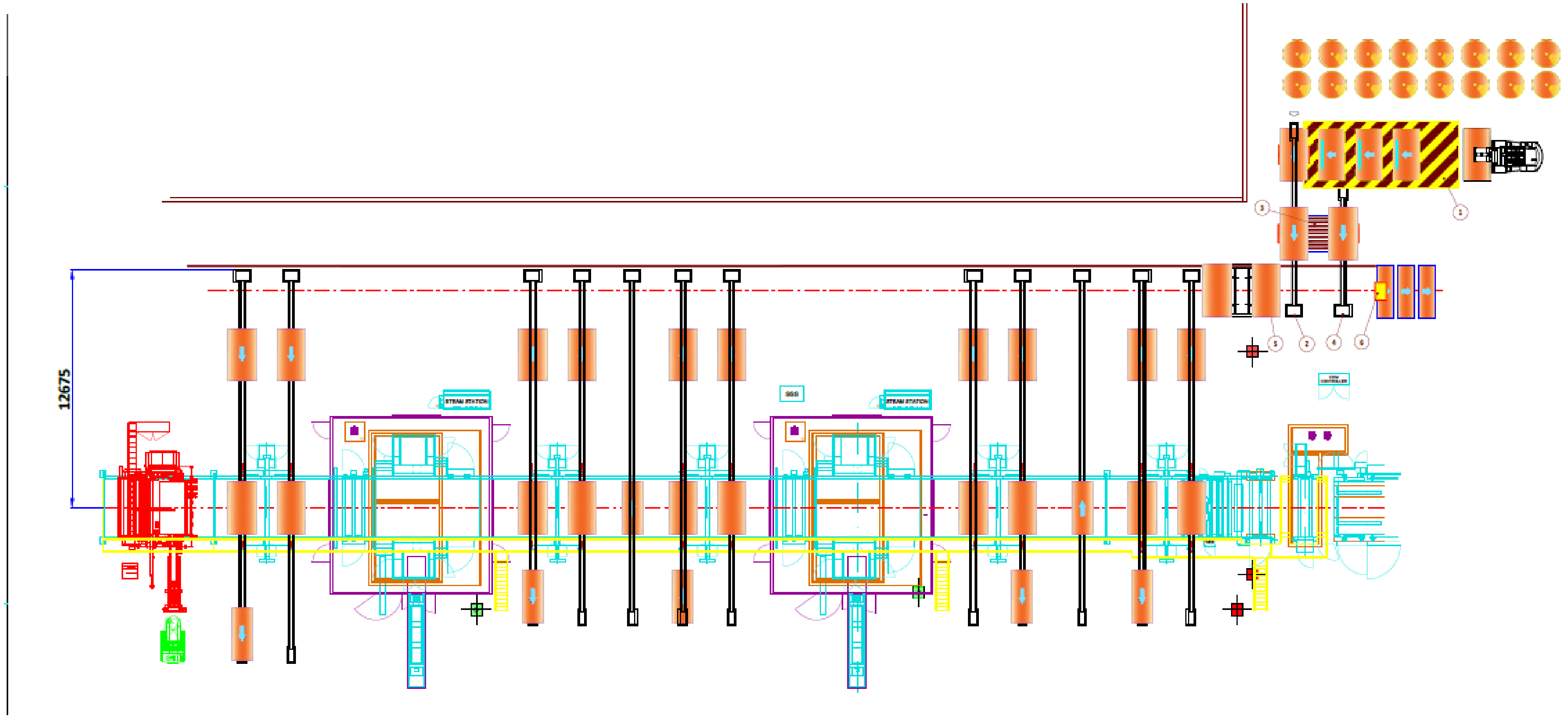
“TP” Project



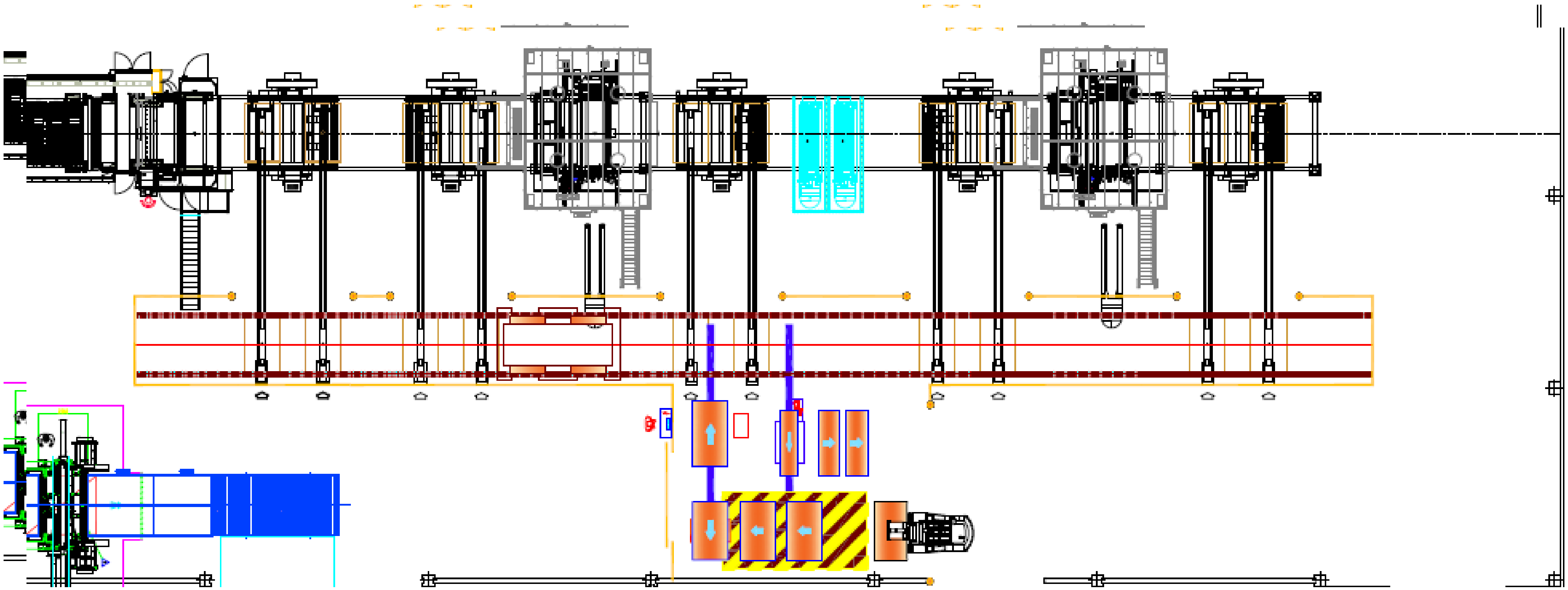
M1240SDR

Item	Qtá.	Descrizione
1	1	Rampa con scorta di 3 bobine
2	1	Binario con carro di 10 mt. mm con espulsore
3	1	Trasportatore Pneumatico
4	1	Binario di carro di 7 mt. mm con centratore
5	1	Navetta Transfer-Car Doppia sul Pavimento
6	1	Espulsore idraulico

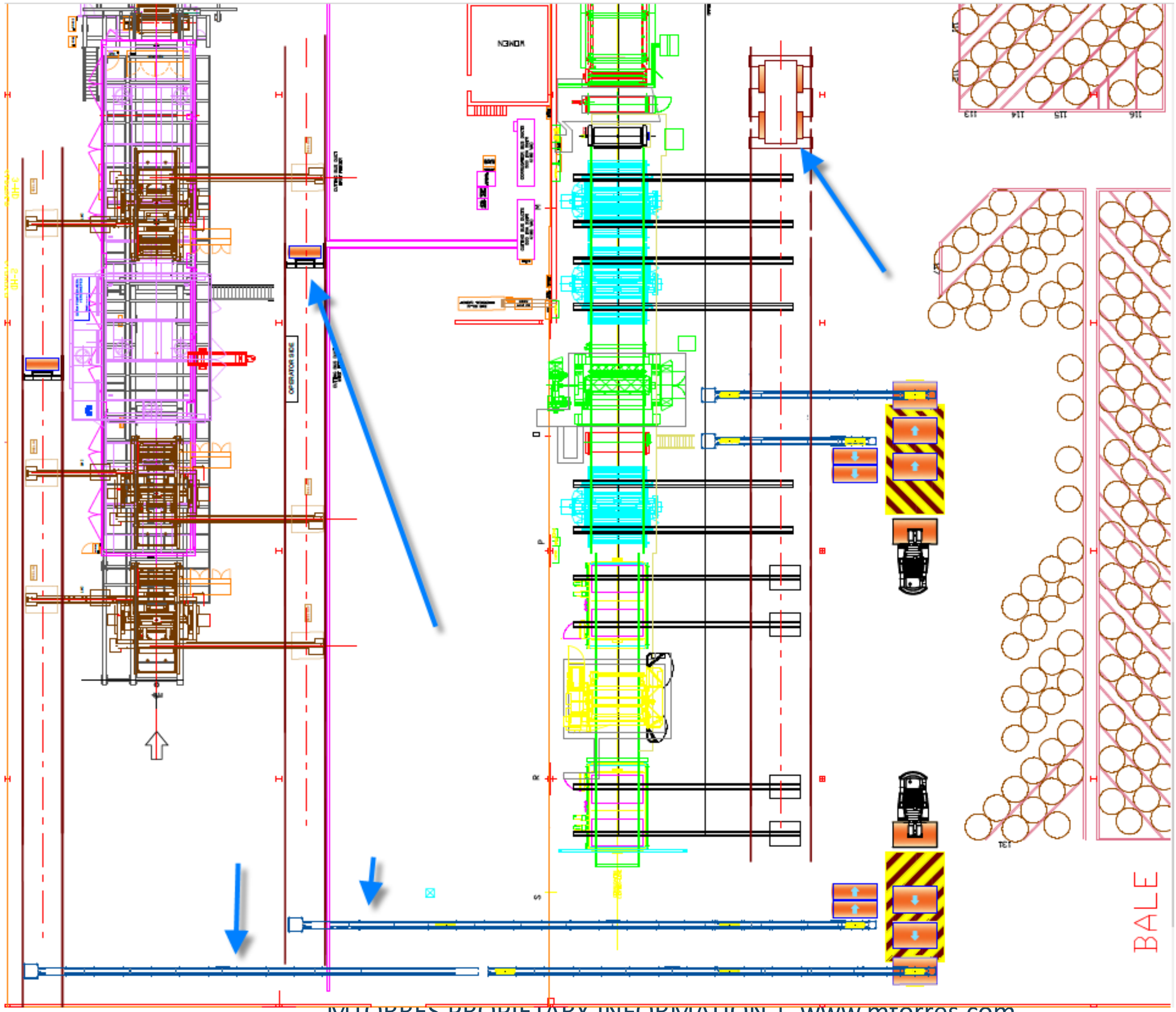
M1240SDR



CME1213



STLS



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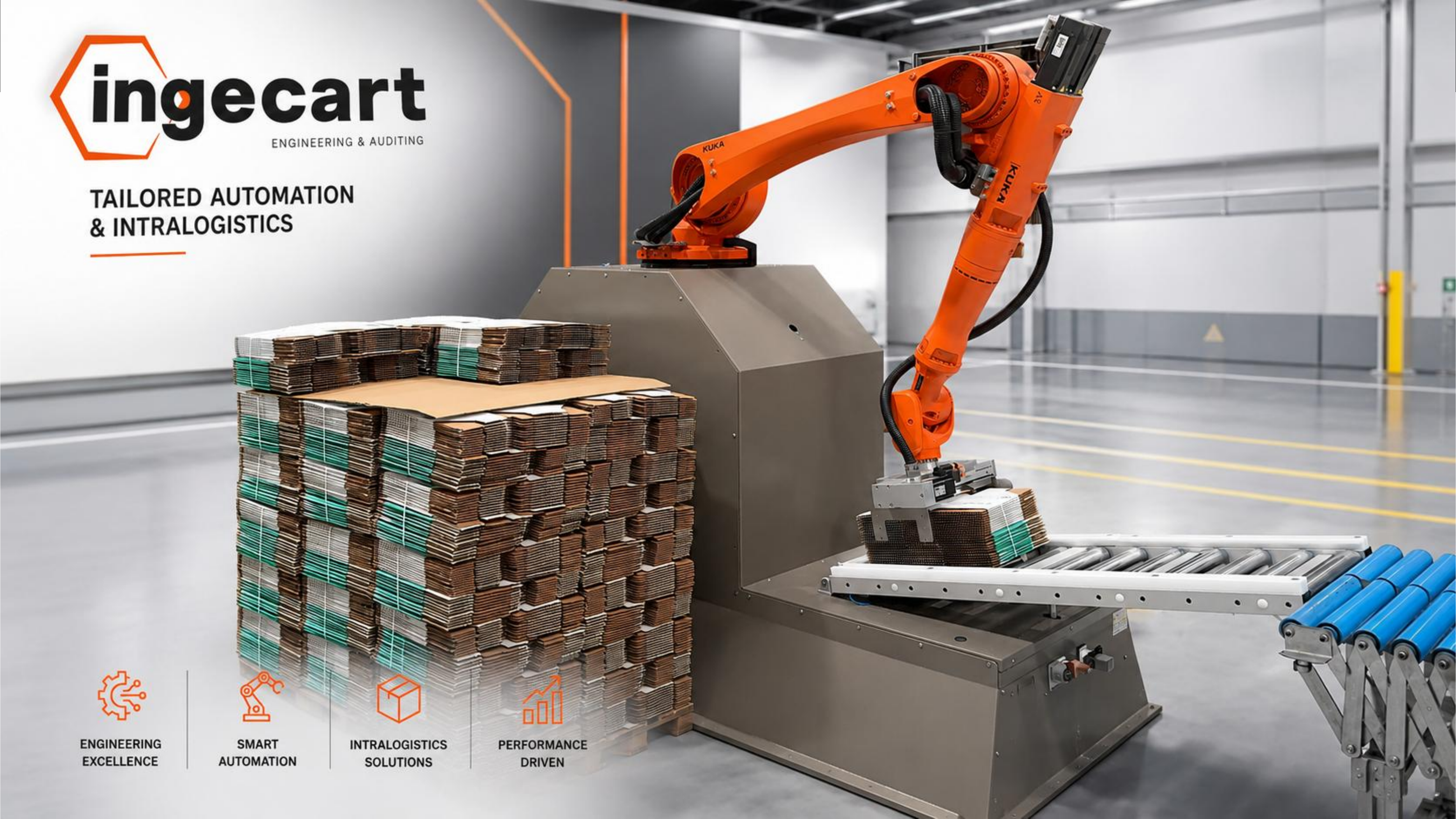
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KUKA
KUKA
ENGINEERING
EXCELLENCE
SMART
AUTOMATION
INTRALOGISTICS
SOLUTIONS
PERFORMANCE
DRIVEN

Plug&Play Easypack+Palletizer : Stack higher. Stack faster. Zero errors..

12 stable production rate with interlayers, automatic pallet exit

Up to 1200×1200×2300 palletizing area

From 250×250 to 700×900 bundle size

Compact design: 3×3 meter footprint

🔌 No production stoppage. Plug, configure, and palletize.

🛠️ Universal gripper. 50+ sizes. Zero changeover time.

No tools needed. Just a recipe.

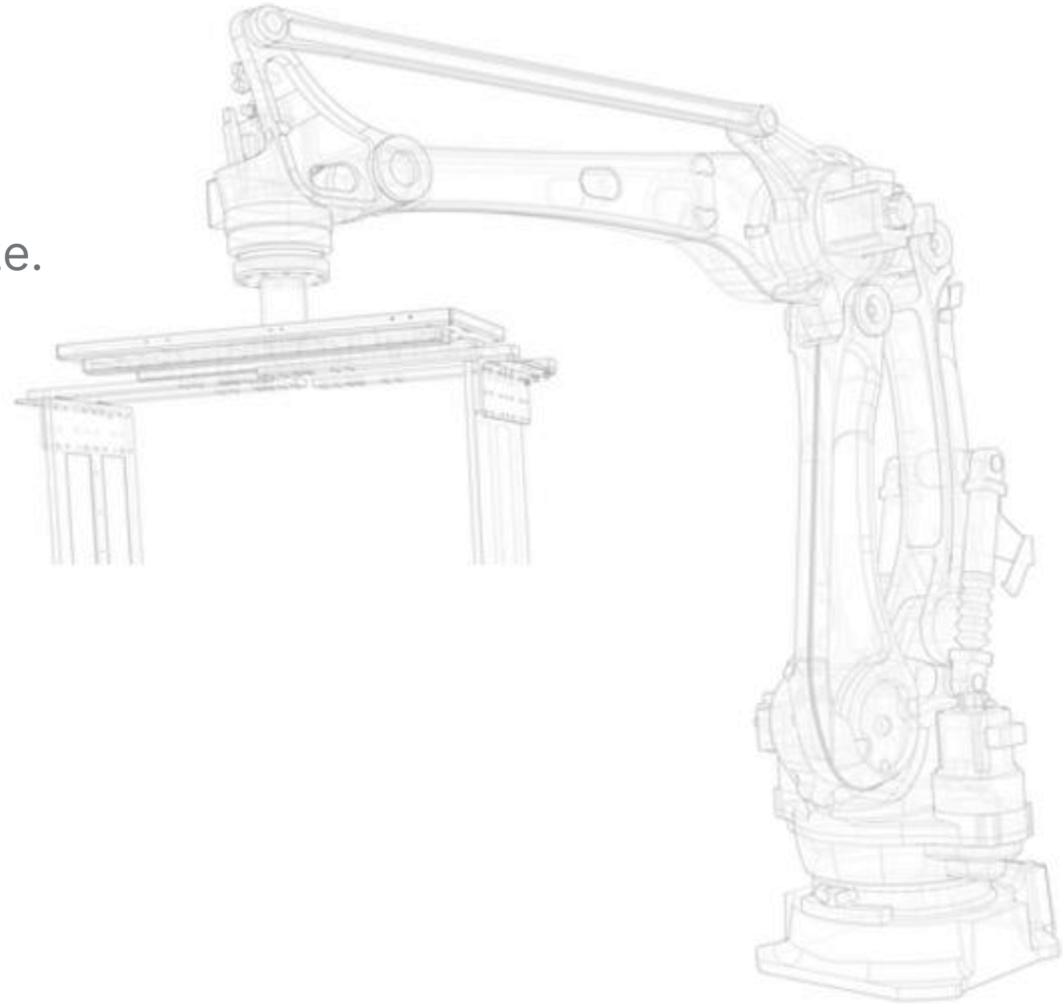
📈 ROI: Automation pays for itself.

🧠 Multi-language HMI + ERP. Real-time data.

Traceability and control from day one.

📦 Scale without redesigning your plant.

Growth doesn't have to be complicated.



Custom palletizer: Stack higher. Stack faster. Zero errors..

13,2 stable production rate with interlayers, automatic pallet exit

Squaring System

Up to 1600×1600×2300 palletizing area

From 350×400 to 1400×1500 bundle size

🛠️ Universal gripper. 50+ sizes. Zero changeover time.

No tools needed. Just a recipe.

📈 ROI: Automation pays for itself.

🧠 Multi-language HMI + ERP. Real-time data.

Traceability and control from day one.

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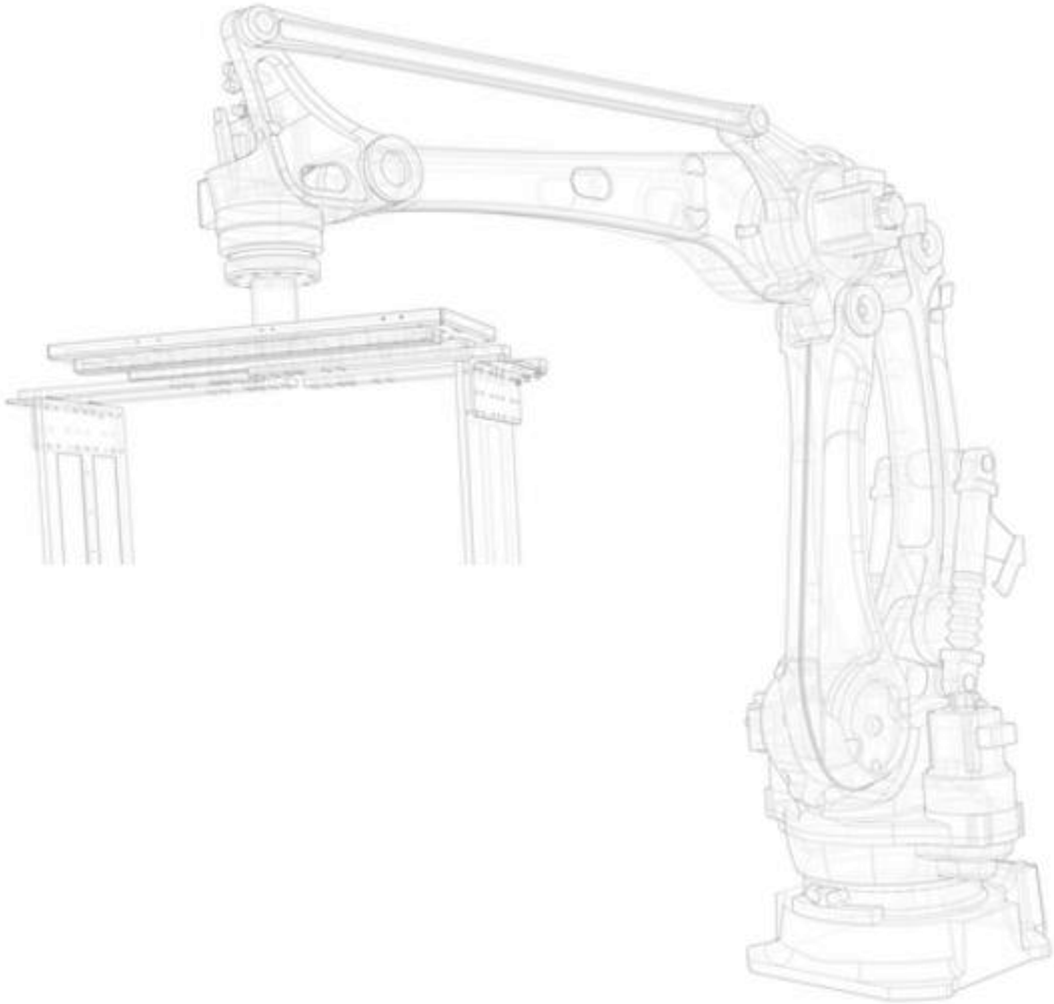

Custom High Speed Palletizer

Las máquinas de conversión de alta velocidad que producen a ritmos de más de 400 piezas por minuto o con múltiples salidas requieren formadores de carga rápidos y eficientes.

Nuestros paletizadores robóticos y sistemas transportadores ofrecen soluciones a medida para cada salida de máquinas FFG y RDC

Robot Arm Stacking Ability: (Pallet size: 1200W*1200L, 4 Bundles/Layer; (485W*635L*125H)
High Speed Single Bundle Max. Stacking Time: 23±1times/Min。

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INTRALOGISTICS
AMR

INTRALOGISTICS
AMR

Supported by KUKA, Ingecart has develop a unique automated system for improving the WIP management, efficiency precision cost reduction .

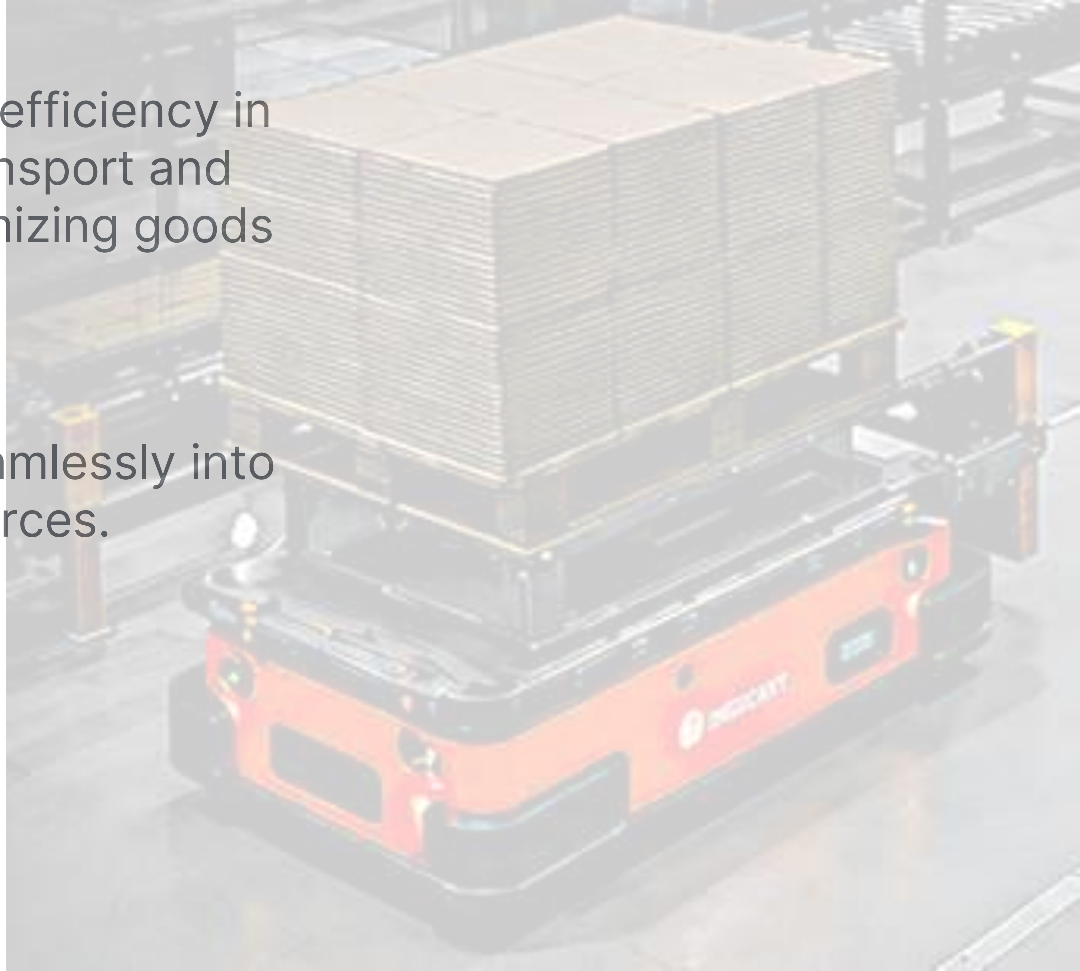
Ingecart Intralogistics Management System
AMR-Based Solution

Ingecart Intralogistics Management System – AMR-Based Solution

Automate. Optimize. Maximize Every Square Meter.

Autonomous Mobile Robots (AMRs) significantly increase efficiency in storage and production environments. They automate transport and logistics processes, accelerating task execution and optimizing goods flow.

Ingecart's compact and flexible AMR models integrate seamlessly into existing supply chains – saving substantial time and resources.



Ingecart Intralogistics Management System
AMR-Based Solution

Fully automated movement of loads, WIP, and finished goods between warehouse, production, and converting machines

Real-time communication with production planning systems

📦 Logistics Efficiency

Just-in-time delivery of materials to converting stations

Reduced waiting times and improved overall equipment effectiveness (OEE)

Synchronized material flow aligned with production pace

📐 Maximum Space Utilization (WIP per m^2)

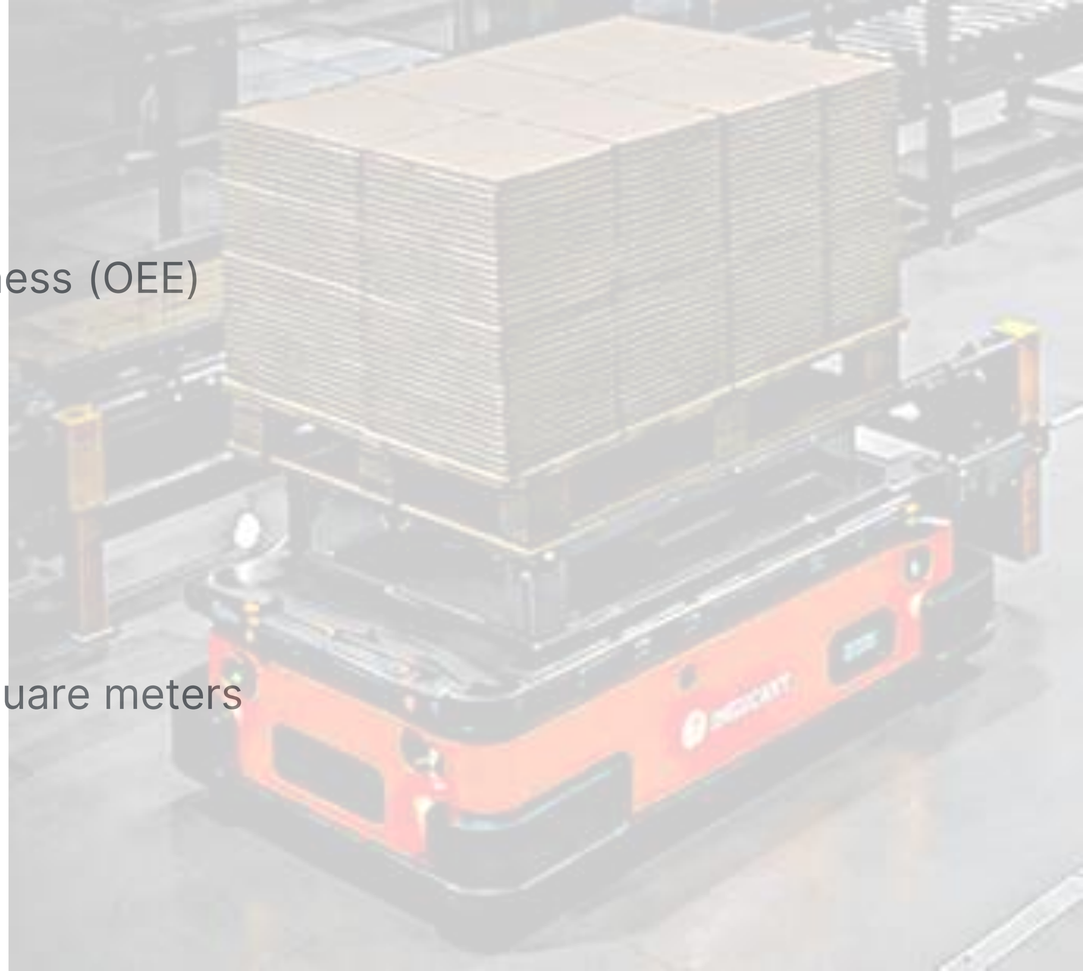
Compact AMR design navigates narrow aisles and tight spaces

Optimized WIP placement in minimum footprint

Intelligent routing allows higher storage density without adding square meters

🔁 Flow Management Between WIP & Converting Machines

Dynamic management of work-in-progress buffers



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ENGINEERING & AUDITING

SR-1400
WASTE SYSTEM

SR-1400

The complete waste collection & transport system.

One system. All waste.

Casemakers | Die-cutters |
Corrugator | Sheet waste

LOW ENERGY

LOW NOISE

NO DUST

FIRE RISK REDUCED

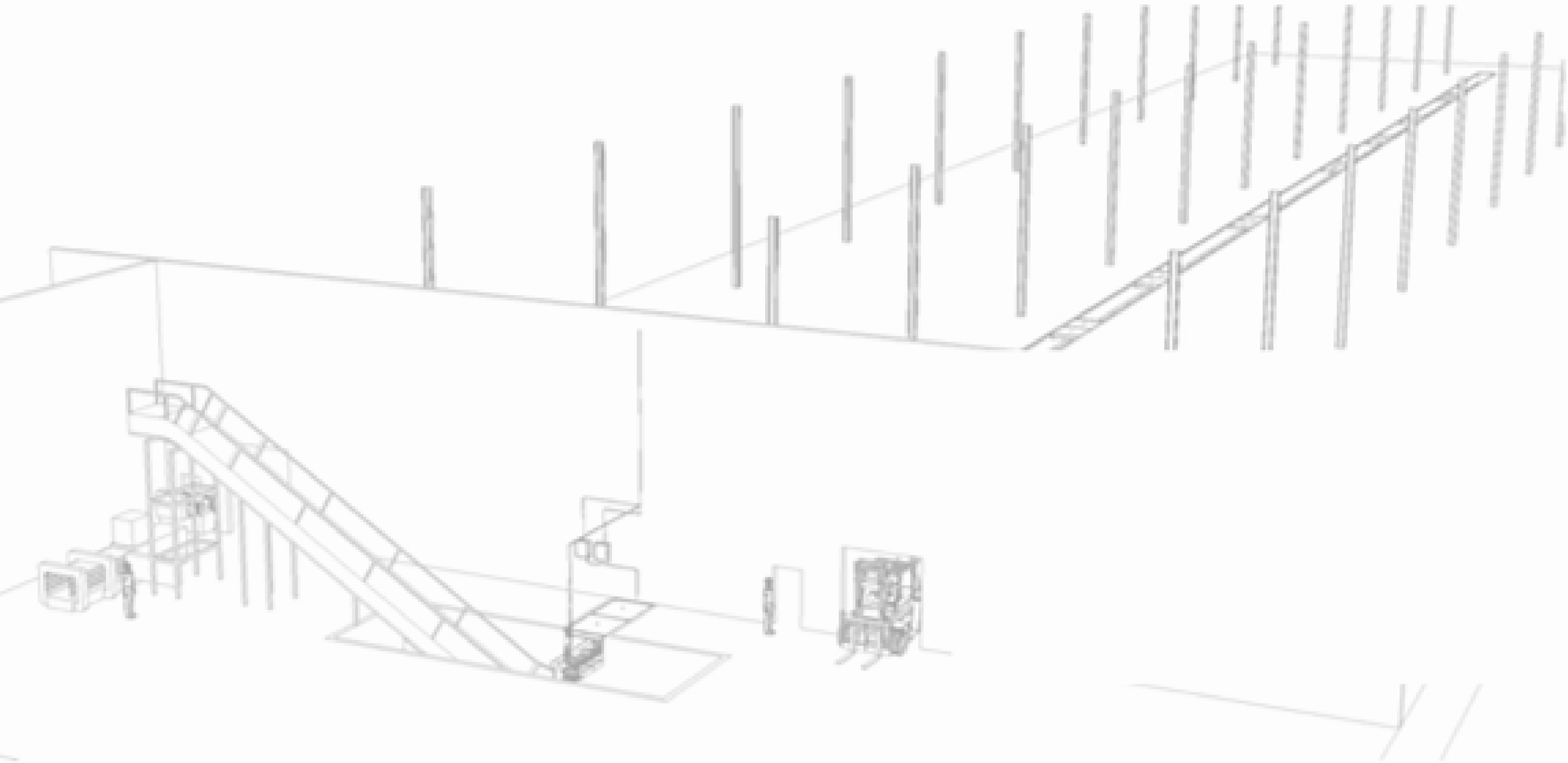
SR-1400
unified solution for collecting and transporting all waste

WHAT IT HANDLES

- Casemaker waste
- Die-cutter waste
- Corrugator waste
- Sheet waste

KEY BENEFITS

- Very low energy consumption
- Low noise level
- No airborne dust
- Reduced fire hazard



SR-1400

unified solution for collecting and transporting all waste

Auto-lubrication system

Sectional stop at each machine:

Safety push-buttons allow operators to stop non-operative sections, improving safety and saving energy.

Anti-wear plates at line ends

Special chain & arm design

Developed after studying multiple customer installations. Eliminates common fixing failures and constant maintenance issues.

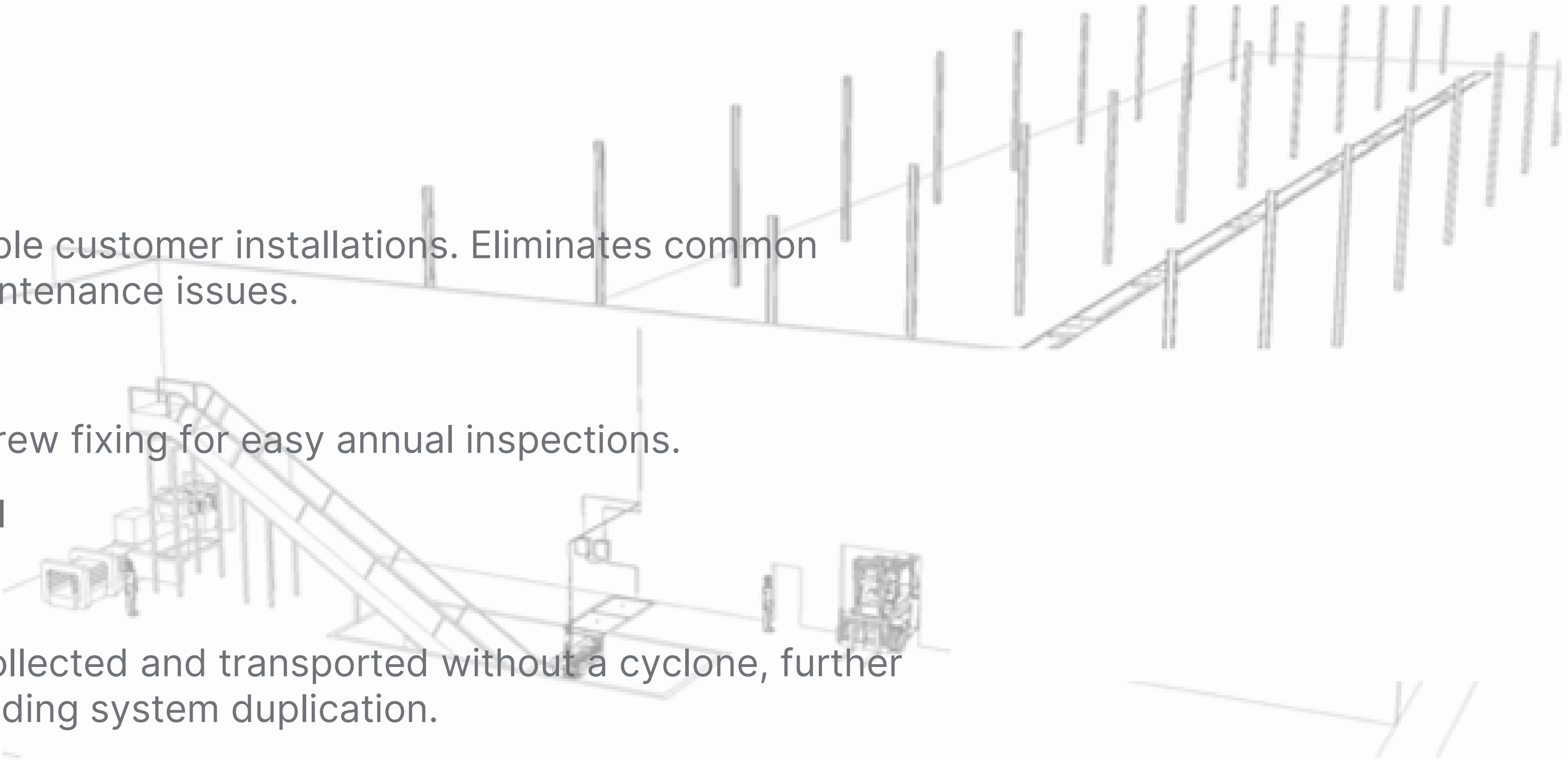
Chain link breaking strength

35,000 kg – Metal fins allow screw fixing for easy annual inspections.

Speed inverters for flow control

Complete system integration

Waste from the corrugator is collected and transported without a cyclone, further reducing energy costs and avoiding system duplication.



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ENGINEERING & AUDITING

RFID
ROLL
MANAGEMENT

fully integrated, cost-effective RFID-based solution to automatically identify, track, and validate paper rolls at every critical station

1 ROLL RECEIVING
Rolls are received and RFID tags are encoded with roll data.

2 WAREHOUSE STORAGE
Rolls are stored in the warehouse and tracked in real-time. System knows exact location and inventory.

3 ROLL STAGING
Rolls are selected and staged for production based on planned orders (MES/ERP).

4 CORRUGATOR LOADING (LEFT LANE)
Roll is scanned and validated at the loading lane. The roll is placed on a motorized tray that moves automatically into position.

5 CORRUGATOR LOADING (RIGHT LANE)
The roll is delivered to the corrugator infeed. System verifies roll ID, grade, width, and quality.

6 CONSUMPTION TRACKING
Real-time consumption data is updated in MES: meters used, weight used, and estimated remaining.

7 CORE RETURN
Core is scanned at return. System validates and updates inventory automatically.

REAL-TIME VISIBILITY
Track roll location, status, consumption and core returns in real-time.

ACCURATE DATA
Eliminate manual errors and ensure accurate inventory and reporting.

SMART DECISIONS
Use real data to optimize planning, reduce waste and increase efficiency.

SEAMLESS SYSTEM INTEGRATION

Ingecart

INGECART RFID SYSTEM

BI-DIRECTIONAL DATA EXCHANGE

ERP / WMS / MES SYSTEMS

- Roll master data
- Inventory updates
- Consumption data
- Core returns
- Production reporting

KEY BENEFITS

AUTOMATIC STATION VALIDATION
Each roll is scanned and verified at every critical station.

REAL-TIME CONSUMPTION TRACKING
MES calculates remaining meters and weight automatically.

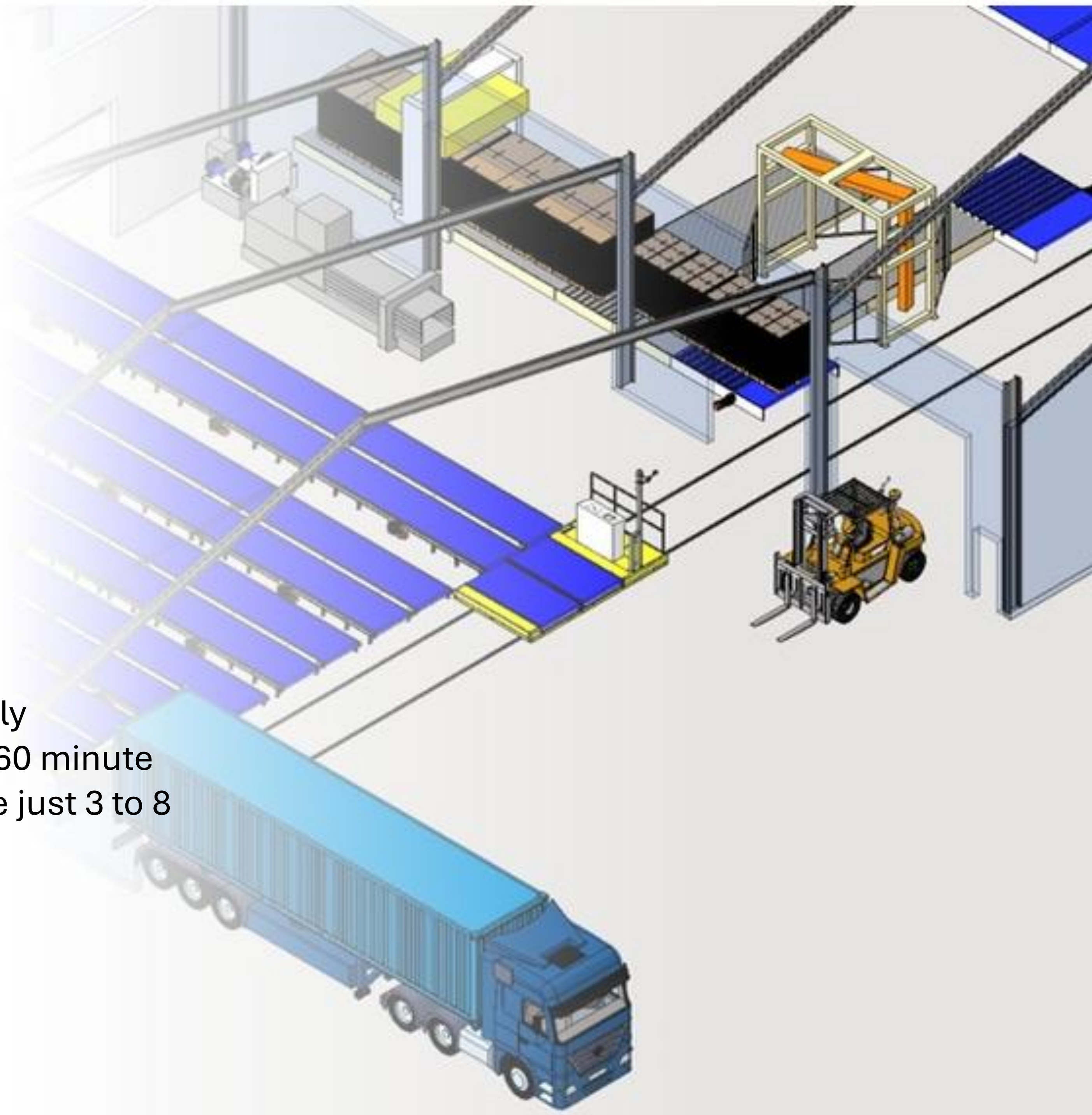
ERROR-PROOF ROLL IDENTIFICATION
Eliminates manual data entry and misidentification.

SEAMLESS ERP INTEGRATION
Bi-directional data exchange with existing systems.

CORE TRACKING & RETURN
RFID enables automatic core reconciliation and inventory updates.

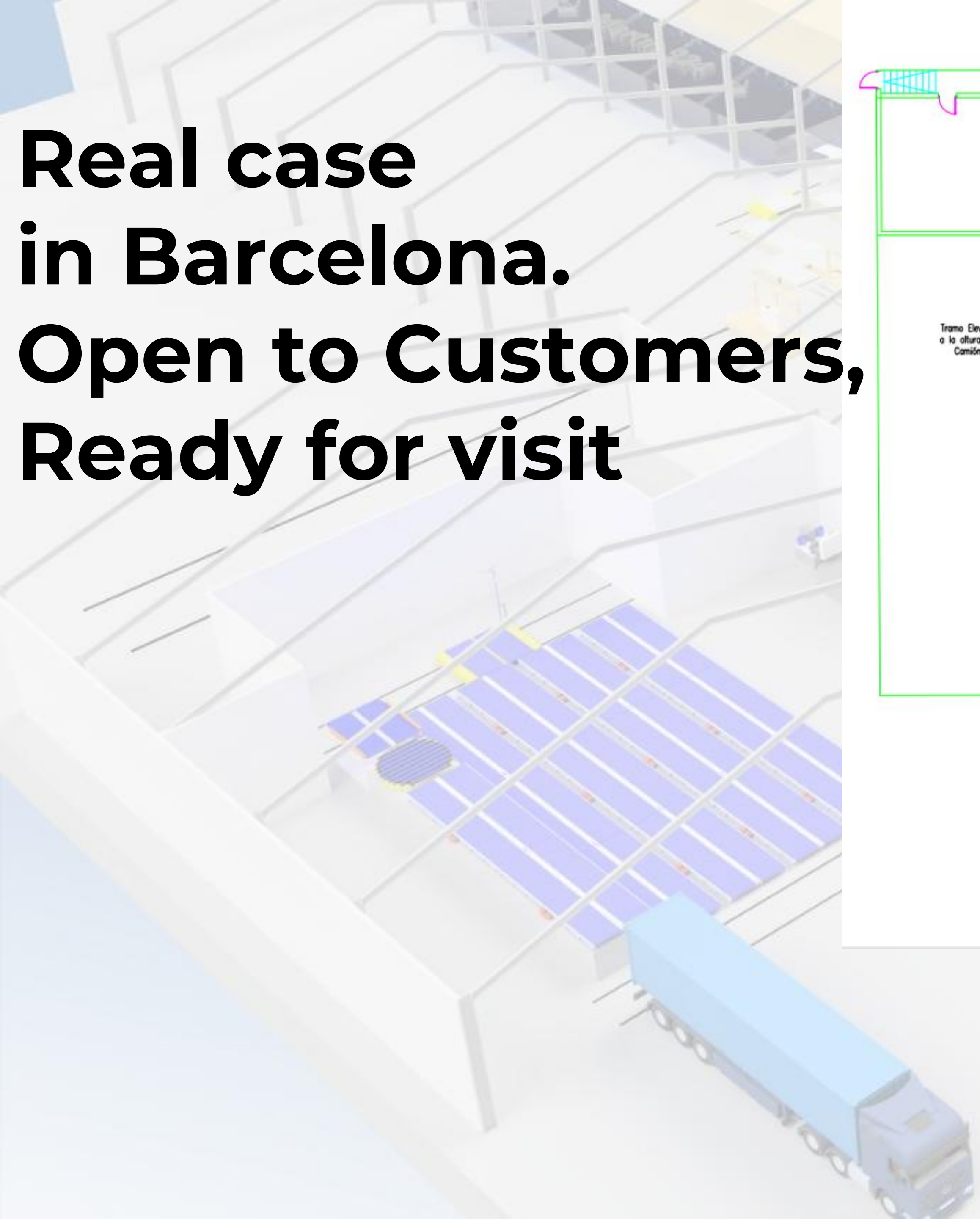
REDUCED FORKLIFT TRAFFIC
Planned roll staging minimizes unnecessary movements.

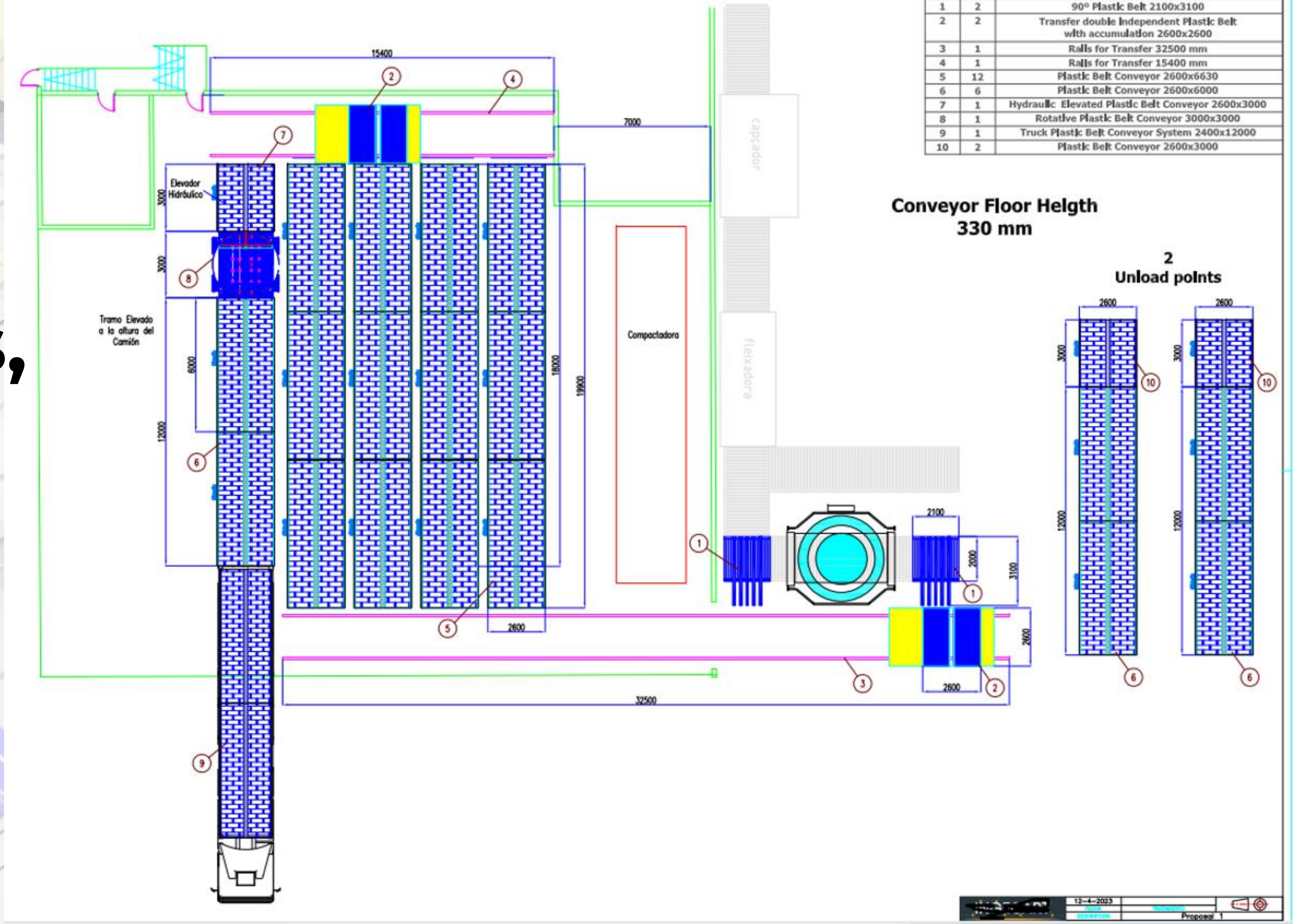
AUTOMATIC TRUCK LOADING



Automatic Truck Loading Systems (ATLS) completely revolutionize warehouse logistics by turning 30-to-60 minute manual loading processes into operations that take just 3 to 8 minutes.

Real case in Barcelona. Open to Customers, Ready for visit





1	2	90° Plastic Belt 2100x3100
2	2	Transfer double Independent Plastic Belt with accumulation 2600x2600
3	1	Rails for Transfer 32500 mm
4	1	Rails for Transfer 15400 mm
5	12	Plastic Belt Conveyor 2600x6630
6	6	Plastic Belt Conveyor 2600x6000
7	1	Hydraulic Elevated Plastic Belt Conveyor 2600x3000
8	1	Rotative Plastic Belt Conveyor 3000x3000
9	1	Truck Plastic Belt Conveyor System 2400x12000
10	2	Plastic Belt Conveyor 2600x3000

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ENGINEERING & AUDITING

DIRECTION & AUDIT

Independent consulting for the corrugated industry

At Ingecart, we understand that corrugated plant managers often stand alone when making critical decisions. The complex technical reality requires advice from highly qualified specialists with no external interests.

Our mission is to be your journey partner. We present objective, impartial reality – analyzing different options together with their technical and organizational impacts.

WE SERVE YOU IN THREE MAIN SCENARIOS:

1 EXISTING FACTORIES

We analyze your equipment and processes.
We present suggestions with clear justification.
We help you evaluate offers for new or used equipment, assist with implementation, support staff training, and follow production improvements.

2 NEW PROJECTS

We help you develop financial viability plans and ROI calculations. Support includes budget development, engineering and technical studies, machinery purchase negotiation, risk management, training, and complete project follow-up from commissioning to new goal definition.

OBJECTIVE ANALYSIS | INDEPENDENT JUDGMENT | TECHNICAL & ORGANIZATIONAL IMPACT | IMPLEMENTATION SUPPORT | TRAINING & DEVELOPMENT | CONTINUOUS IMPROVEMENT

CONSULTANCY SERVICES

Our consultancy service is a fundamental part of our offering. We specialize in expert advice and customized solutions to help optimize processes and achieve business objectives.

In our consultancy meetings, we adopt a collaborative approach:

- Listen carefully to concerns and objectives
- Examine existing processes and operations in detail
- Identify inefficiencies, bottlenecks, and improvement points
- Work closely with your team to analyze findings
- Discuss possible solutions, emerging technologies, and industry best practices
- Provide practical, actionable recommendations

WHY INGECART?

- Complete independence from manufacturers and equipment vendors
- Deep experience in paper, corrugated board, and production
- Unbiased advice focused entirely on your success
- End-to-end service: from initial audit to project completion

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ENGINEERING & AUDITING

TAILORED AUTOMATION
& INTRALOGISTICS

The Corrugated Plant of the Future Starts Here

ENGINEERING EXCELLENCE

SMART AUTOMATION

INTRALOGISTICS SOLUTIONS

PERFORMANCE DRIVEN